

Date Night Is a Fragile Institution: Bernoulli Flakes, Hard Quorums, and the Inspection Paradox of Saturday Night

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Abstract

Wanting one other person—a date, a catch-up, “quality time”—is a statistically lonely desire. Each invitee shows independently with probability p ; the gathering happens iff at least q people arrive. A person is not alone iff they show *and* the gathering happens. At $q = 2$ and $p = 0.70$, the closed-form chance of a night alone is 0.510 for a dyad ($k = 2$) versus 0.300 for a pub ($k = 24$): extra isolation 0.210 above the own-flake floor $1 - p = 0.300$. The $q = 1$ kill is exact zero: if showing up to an empty table counts, dyads and pubs are identical. Monte Carlo on 2400 numpy people $\times$ 30 nights $\times$ 5 seeds matches ($\Delta = 0.212 \pm 0.003$). Replacing pubs with dyads, same person-slots, raises population nights-alone by 0.210. In a 50/50 mix, most *events* are dates (mean proposed size 3.7) but the attendance-weighted feed is a pub (mean 10.865, ratio 2.943). Airline-overbooking a date to 6 people approaches the flake floor and ceases to be a date. This is a generative cartoon, not a UCLA survey and not a measurement of the loneliness epidemic.

1 Laugh, then think

The laugh: the statistically honest cure for a cancelled Friday is to triple-book the other person, or to go where cancellation is impossible—the pub. The think: intimacy and flake-robustness are binomial-incompatible under independent shows. A society that moves social life from large, low-quorum rooms (pubs, churches, clubs) into one-on-one catch-ups can raise nights-alone *without anyone becoming less willing to show up*.

Isolation is associated with mortality risk [1]. Loneliness is not the same thing as isolation: it is a subjective gap between desired and achieved contact [2–4]. We do not collect those scales. We measure a mechanical source of the gap: nights alone, and a Saturday-night feed that over-represents surviving crowds.

2 What this is not

Feld’s friendship paradox—your friends have more friends than you do—is length-biased sampling on degree [5, 6]. So are the happiness paradox [7] and the majority illusion [8]. Feld already noted that beaches feel more crowded than they “usually” are. We do not retitle that.

Group-chat folklore already knows that flakes kill small plans. The identifying pieces here are (i) a closed form, (ii) a $q = 1$ kill that isolates quorum from “people got flakier,” (iii) a quality-over-quantity comparative static at fixed person-slots, (iv) survivorship plus inspection of the feed, and (v) a copula that holds the marginal show-up rate fixed.

Granovetter thresholds make attendance a strategic complement: you go if enough others go [9]. Our people are worse and simpler. They flip independent coins. Small gatherings still collapse.

Social-media FOMO is a human measurement [10]. Our “feed” is just happening events, weighted by how many people were there to post.

3 Model

k invitees, including you. Each shows independently with probability p . Quorum q . You attend a happening event iff you show and at least $q - 1$ of the others show:

$$P(\text{attend}) = \begin{cases} p & q \leq 1, \\ p \cdot P(\text{Bin}(k-1, p) \geq q-1) & q \geq 2. \end{cases} \quad (1)$$

$P(\text{alone}) = 1 - P(\text{attend})$. The own-flake floor is $1 - p$: a stadium cannot socialize a person who stays home. Extra isolation above that floor is $p \cdot P(\text{Bin}(k-1, p) < q-1)$. For the pair quorum $q = 2$ this is exactly $p(1 - p)^{k-1}$: everyone else flaked.

H1. At $q = 2$ and $p \in (0, 1)$, $\Delta_{\text{alone}} = P(\text{alone} \mid k = 2) - P(\text{alone} \mid k = 24) > 0$.

Kill H1. $q = 1 \Rightarrow \Delta_{\text{alone}} = 0$.

H2/H3. In a mixed calendar of dyads and pubs, cancelled dates vanish and remaining person-nights length-bias toward pubs, so the attendance-weighted feed exceeds the event-weighted proposed mean.

Policy. Convert every pub into dyads (same slots): population $P(\text{alone})$ rises. Over-invite a dyad until it matches the pub: it is no longer a dyad.

4 Protocol

Exact binomial identities for $k \in \{2, 3, 4, 6, 8, 12, 24, 48\}$, $p \in \{0.5, 0.6, 0.7, 0.8\}$, quorum rules $\{\text{one}, \text{pair}, \text{half}\}$. Finite-population calendars: 2400 people, 30 nights, 5 seeds, no double booking. Headline cell: $p = 0.70$, $q = 2$, latent correlation $\rho = 0$. Robustness: Gaussian copula $\rho = 0.5$ with Bernoulli(p) margins (unlike a logit shock, this does not move $E[p]$). JSON: `results/exp10_quorum_lonely.json` (120 cells ok, 0 failed, 0.7 s). No hand-edited numbers.

5 Results

H1, exact. Figure 1 plots $P(\text{alone})$ against k . At $q = 2$, $p = 0.70$, a dyad is alone with probability 0.510, a four-person dinner 0.319, a pub 0.300. Extra isolation is 0.210 versus 0.000. The dinner is almost a pub; the date is not. Monte Carlo: $\Delta = 0.212 \pm 0.003$ (dyad 0.511, pub 0.299). Dyad cancellation rate 0.511; pub cancellation 0.000.

Kill. $q = 1$: exact $\Delta = 0.000$ (both equal the flake floor 0.300). MC $\Delta = -0.001 \pm 0.001$. If an empty table counts as company, wanting a date is no lonelier than wanting a crowd. The loneliness of the dyad is the refusal to count that table.

Quality-over-quantity. Turning pubs into dyads raises nights-alone by 0.210 exact, 0.209 ± 0.001 MC (Figure 2). Same invitations, same p , more isolation.

The feed (H3). Half the people are assigned dates, half pubs. The *event* calendar is almost all dates (mean proposed size 3.7). Happening events are only slightly larger (4.143, ratio 1.122) because cancelled dates drop out. Weight by attendees—the Instagram of Saturday night—and the mean jumps to 10.865 (ratio 2.943; Figure 3). Stay-homes compare an empty room to a feed of size 10.9. People whose date *happened* still see a feed 8.9 larger than their table. Pub people are 0.500 of the population but 0.587 of person-nights out.

Overbooking. The smallest n that matches the pub within 0.005 is 6 (alone 0.302; Figure 4). Inviting 6 at $m = 3$ gives 0.302. It works. It is not a date.

Correlated Friday nights. Shared latent correlation $\rho = 0.5$ (fixed marginal p) helps dyads: alone 0.445 versus 0.510 independent; pubs stay at 0.303. MC Δ shrinks to 0.140 ± 0.004 but does not die. Independence is the worst case for date night, not the only case.

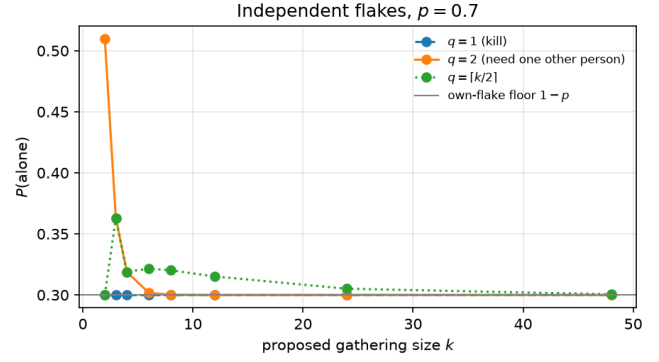


Figure 1: $P(\text{alone})$ versus proposed size. The $q = 1$ curve is the flake floor. Pair quorum makes small k uniquely expensive.

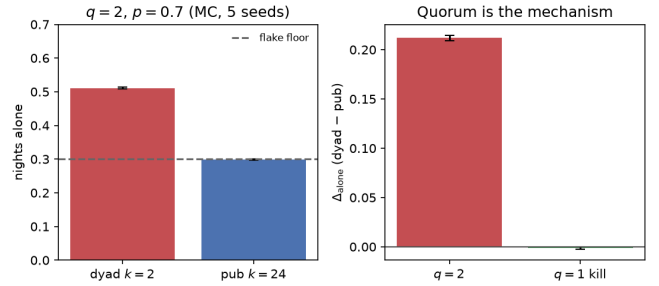


Figure 2: Left: nights alone, dyad versus pub. Right: Δ_{alone} dies at $q = 1$. Error bars are seed sd ($n = 5$).

6 Limitations

Numpy people. No ATUS, no GSS, no IRB, no UCLA items. p is a cartoon of RSVP/flake, not a fitted no-show rate. Real availability is more structured than an equicorrelated probit. Hosts are not strategic; nobody reschedules. We do not claim the loneliness epidemic is “just math,” and we do not recommend triple-booking humans. The result is an identifying DGP: if social contact is implemented as small- k coordination with independent flakes and a hard quorum of two, nights-alone rise relative to large low-quorum rooms *even when willingness to show up is unchanged*.

7 Conclusion

Date night is a fragile institution. Pubs are not. A feed that samples people who went out will look like a pub even when the calendar is dates. The Ig Nobel policy is grotesque on purpose: invite extra people you did not want, or sit where cancellation cannot find you. The serious reading is narrower. “Quality over quantity,” implemented as a shift from flake-robust rooms to dyads, is a binomial tax on contact.

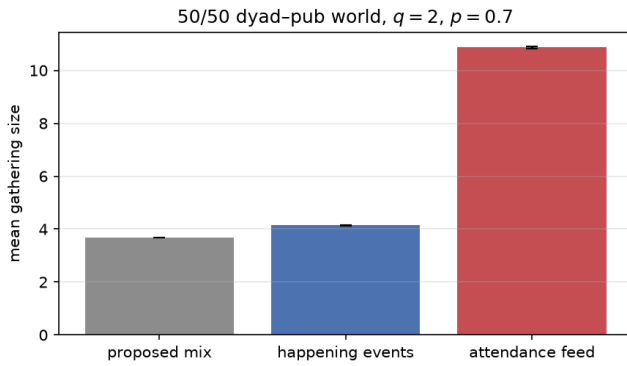


Figure 3: 50/50 dyad-pub world. Most events are small; most attendances are not.

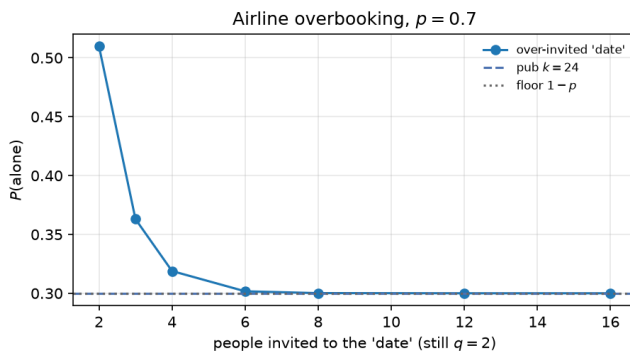


Figure 4: Over-inviting a “date” approaches the pub and the flake floor by ceasing to be a date.

References

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